

CS6P008 - Deep Learning Lab

Lab 8, Date: March 30, 2026

Topic(s): Transfer Learning

Timing: 10:00 AM to 01:00 PM

Max Points: 10

Instructions

1. Provide commented, indented code. Variables should have meaningful names.
 2. Submit two .ipynb files.
The names should be `[student name]_lab[number]_1.ipynb` (before completion of the lab at 1 PM) and `[student name]_lab[number]_2.ipynb` (your final code file by the deadline).
 3. Read the questions carefully before answering. If a question asks to follow a particular approach or to use a specific data structure, then it must be followed.
 4. Write questions in separate text blocks in Jupyter/Colab Notebook before the code blocks containing answers.
 5. All plots should have appropriate axis labels, titles, and legends.
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1. Use a pretrained ResNet18 model (available in `torchvision.models`) for Intel Image Classification (<https://www.kaggle.com/datasets/puneet6060/intel-image-classification>) with the following specifications: (8)
 - Load and preprocess the dataset:
 - Download and extract the dataset.
 - Resize images to 224×224 , normalize pixel values to $[0, 1]$, and convert them into tensors. Also, use standardization.
 - Split the dataset:
 - The dataset contains separate folders for Train and Test. Use Validation data (20% of the training data) to check overfitting/underfitting while training.
 - Create DataLoaders:
 - Implement `train_loader` and `test_loader` using PyTorch `DataLoader`.
 - (**Optional**) Learn to write a Custom data loader to provide greater flexibility in handling custom datasets.
 - Training strategy:
 - Use any standard weight initialization method (only for the newly added layers). Do not initialize the weights for the already existing ResNet layers.
 - Train the model using the AdamW optimizer and Categorical Cross Entropy (CCE) loss.
 - Start with a learning rate = 0.001, and use a learning rate scheduler such that the learning rate is reduced by a factor of 10 if the loss does not decrease for 10 consecutive epochs. Train the model until convergence.

- Evaluation and visualization:
 - Evaluate the model using Accuracy, Precision, Recall, F1-score
 - Compute and display the confusion matrix.
 - Visualize predictions on a few test samples.
 - Use **Weights & Biases (W&B)** to log and visualize:
 - * Training and validation loss
 - * Accuracy, Precision, Recall, F1-score
- 2. For a ResNet18 model trained on the Intel Image Classification dataset, interpret the ROC curve and AUC score for each class, and analyze why the model might achieve high accuracy but low AUC for certain classes, suggesting possible improvements. (2)
- 3. **(Optional)** Explore different methods to visualize a neural network model and the intermediate convolution graphs.